

Chapter 2

Transfer Learning

1. Training from scratch

Means initializing a network with millions of weights randomly & train it purely on your dataset

? When ?

You have a massive, highly unique dataset
(satellite images // complex medical images)

Pros

Model becomes
highly specialized
to your exact dataset

Cons

- Massive datasets
- Enormous compute power

2. Pretrained models & Transfer Learning

Google

Meta

UltraAnalytics

⋮

They trained many models on massive datasets
So they already know basic shapes,
colors, curves ...

2.1 Transfer learning VS. Fine-tuning

Almost All YOLO today are fine-tuning

T.L: Freeze backbone then train the output head (acts as feature extractor)

F.T: We leave the entire network unfrozen allowing gradients to tweak weights

2.2 Freezing Layers

Why?! When we have a small dataset, if we fine-tune the whole network it might forget the general shapes that were learnt at the earlier layers from COCO for eg. this is called

Catastrophic forgetting

model:

